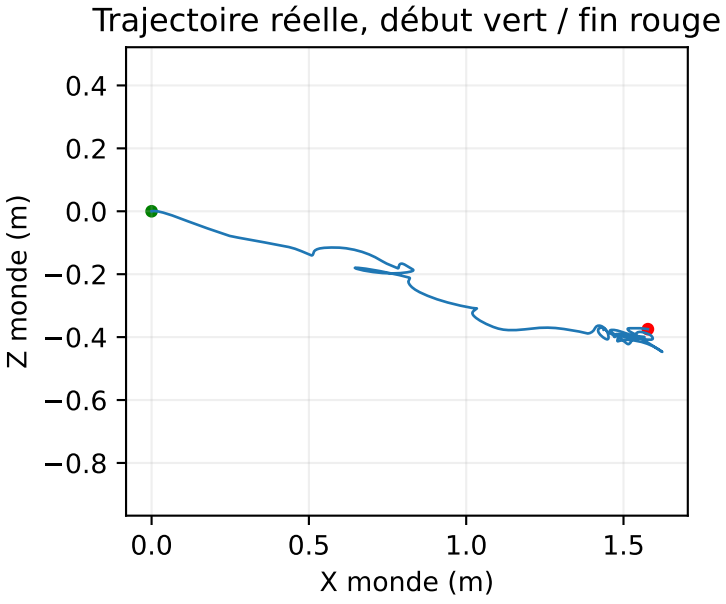
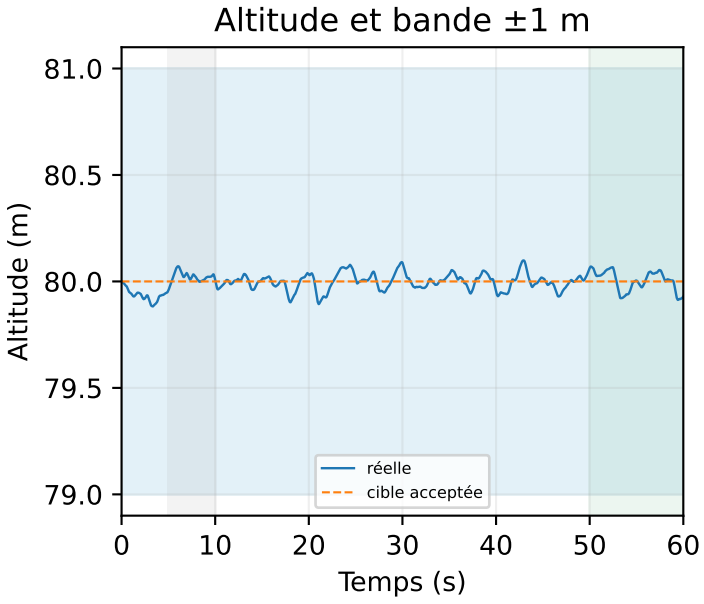
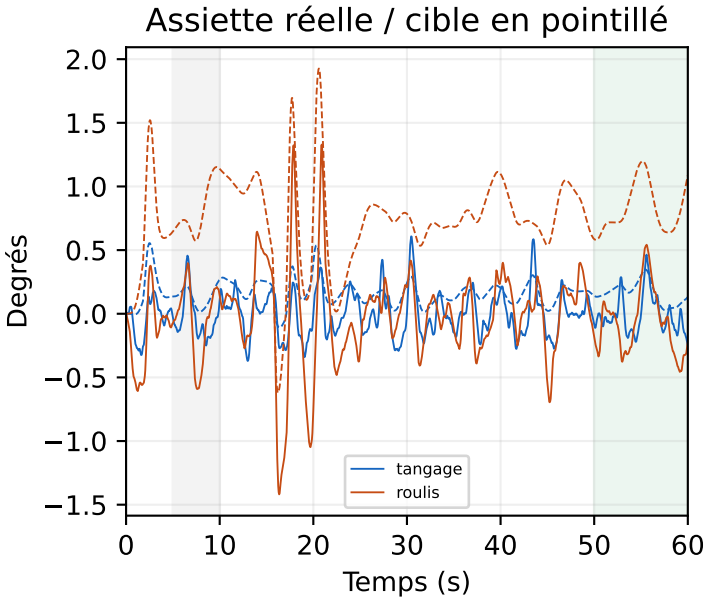
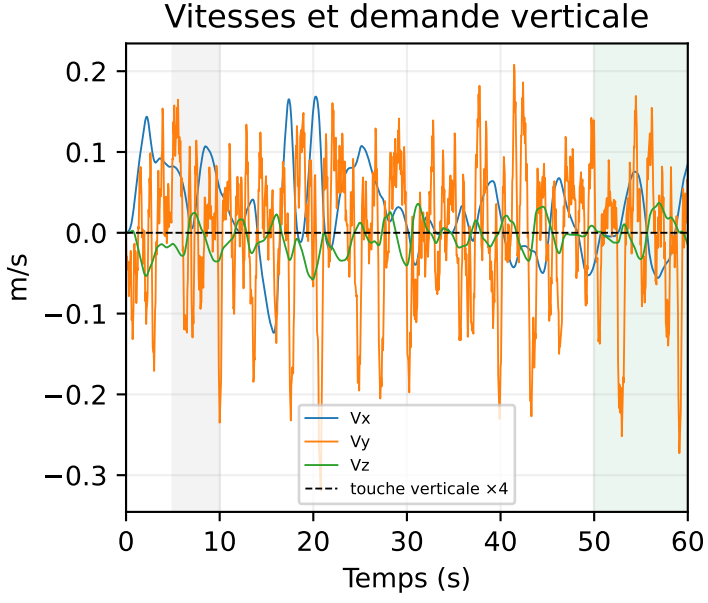
mixed-propellers / limited / hover — PASS

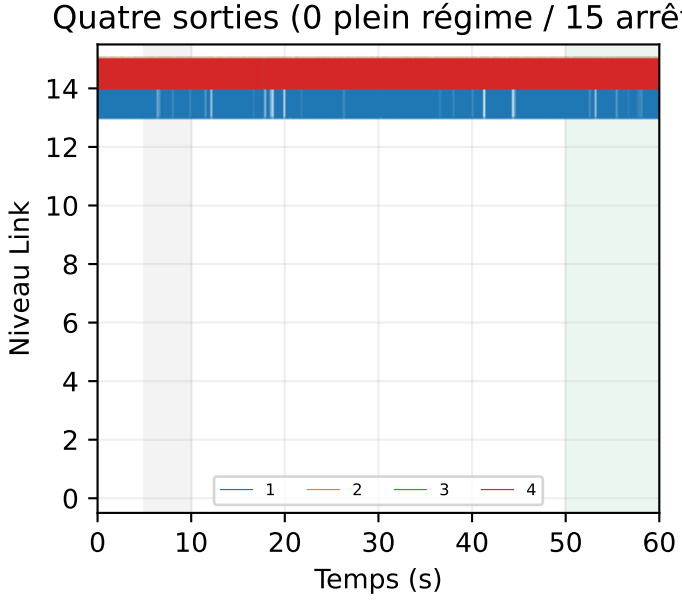
Maintien sans touche pendant toute la simulation.

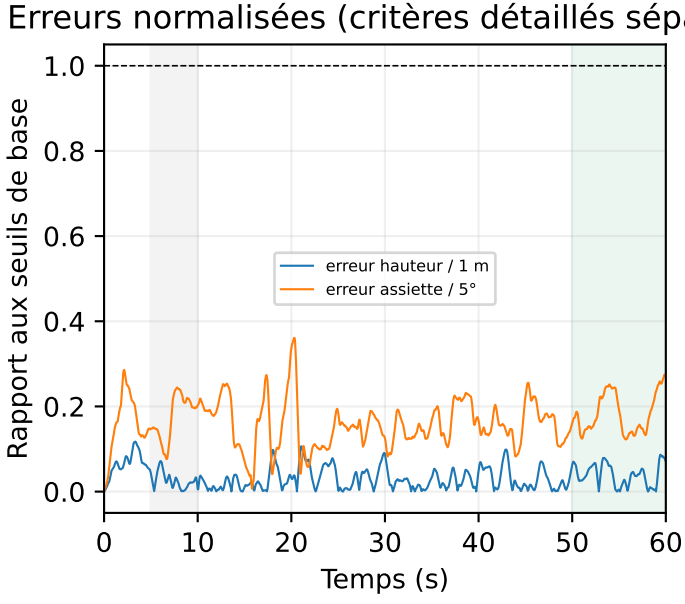










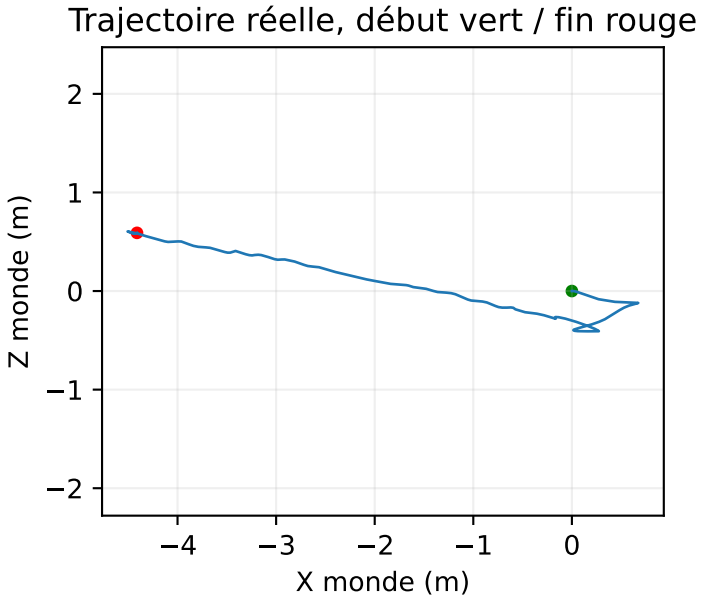


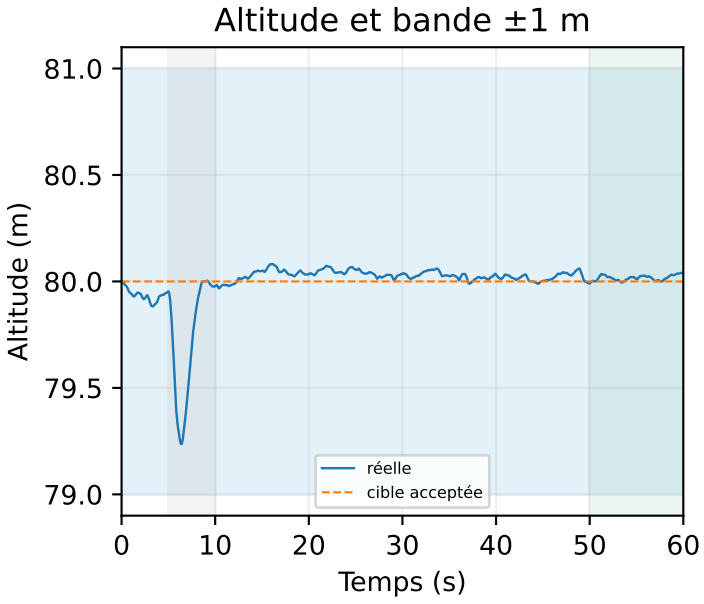
Erreur finale hauteur 0.071 m ; excursion finale 0.157 m ; assiette finale max 0.714° ; freinage — s. Critères échoués : aucun.

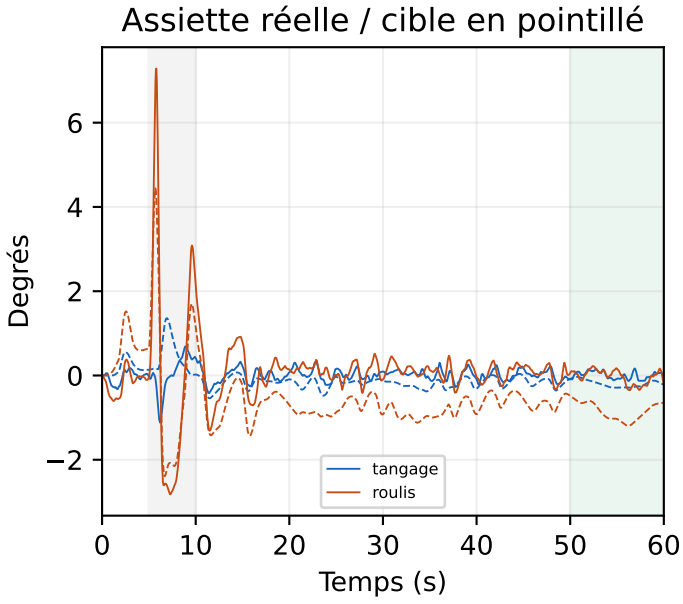
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

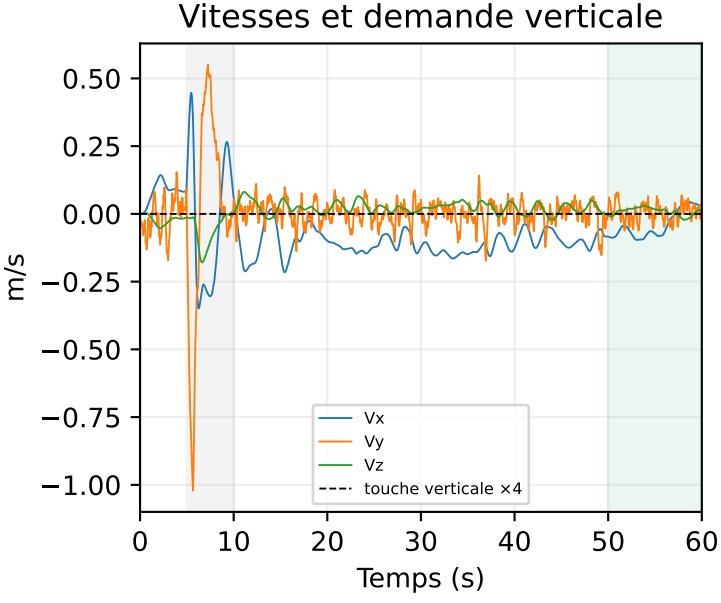
mixed-propellers / limited / load-swing — PASS

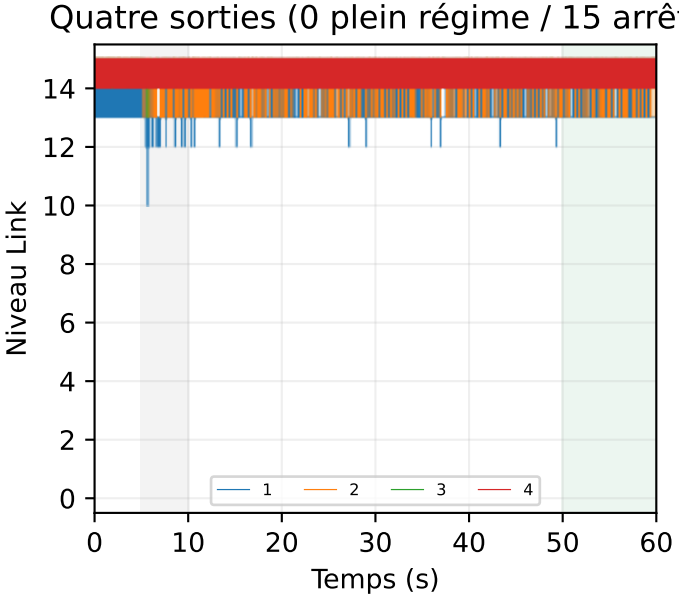
Aucune touche ; charge ponctuelle de 12 kg sur corde 5 m, inclinée initialement de 20°, ajoutée à 5 s.

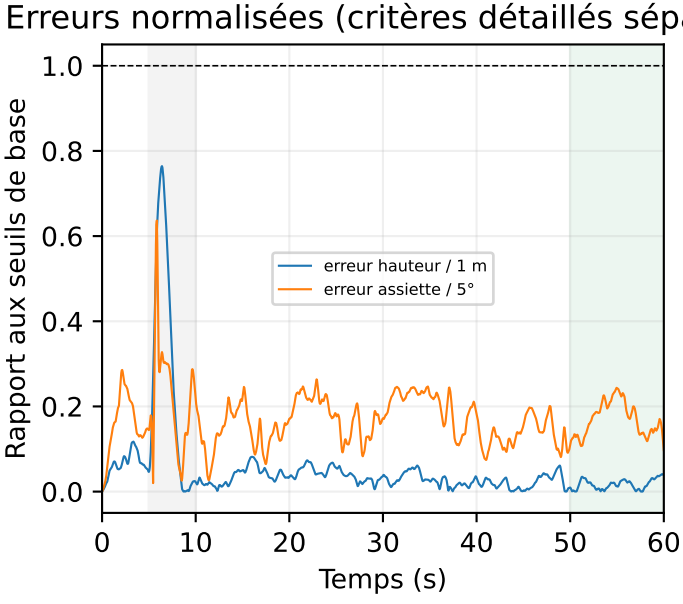










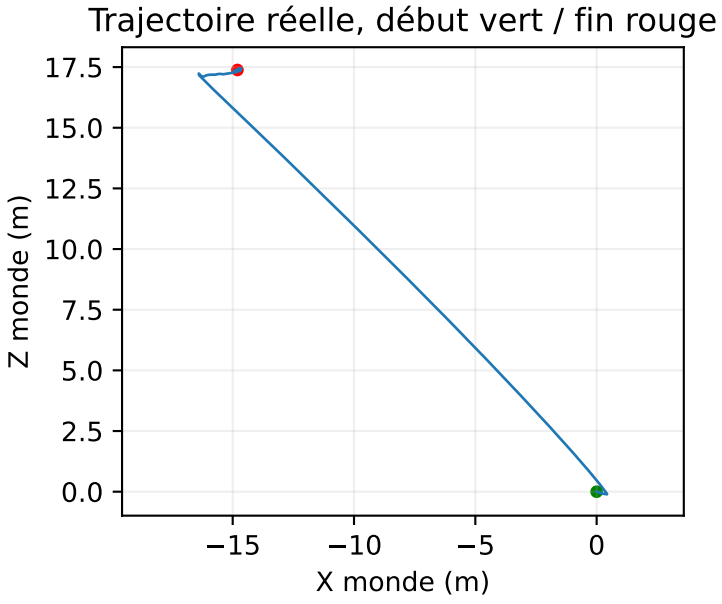


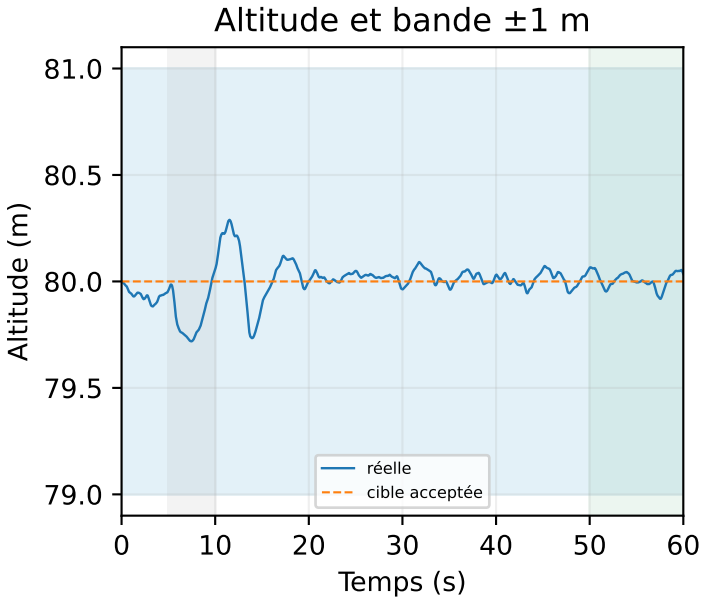
Erreur finale hauteur 0.041 m ; excursion finale 0.049 m ; assiette finale max 0.374° ; freinage — s. Critères échoués : aucun.

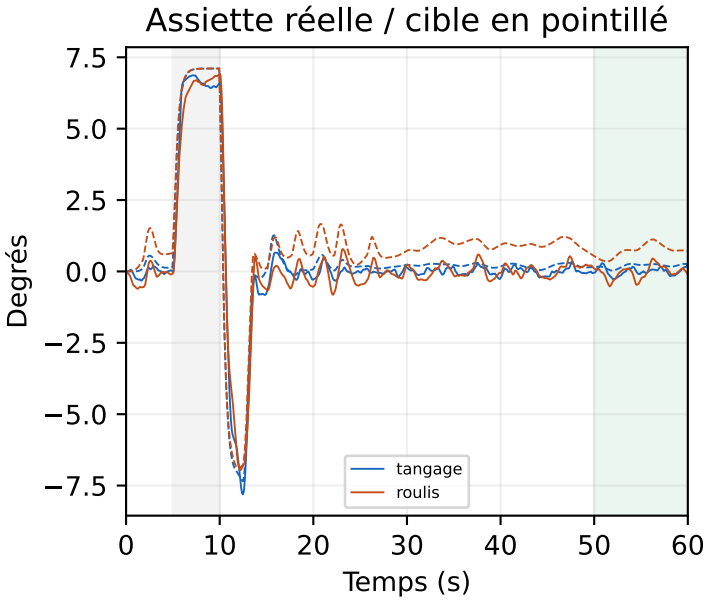
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

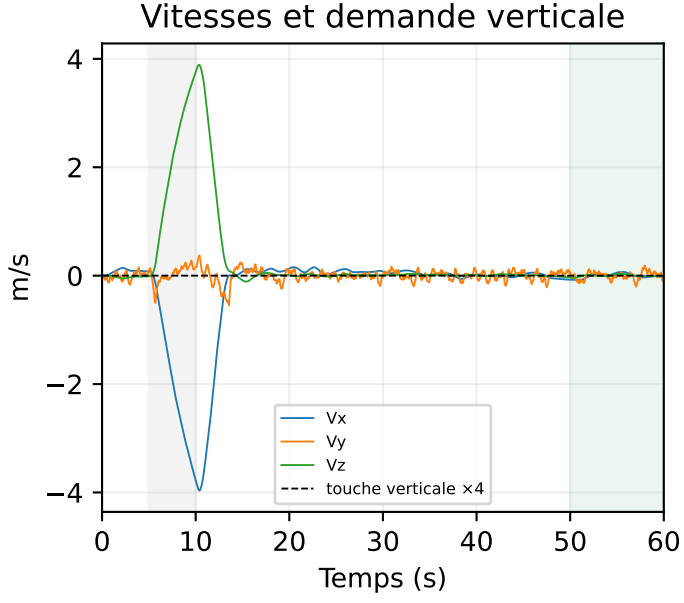
mixed-propellers / limited / forward-left — PASS

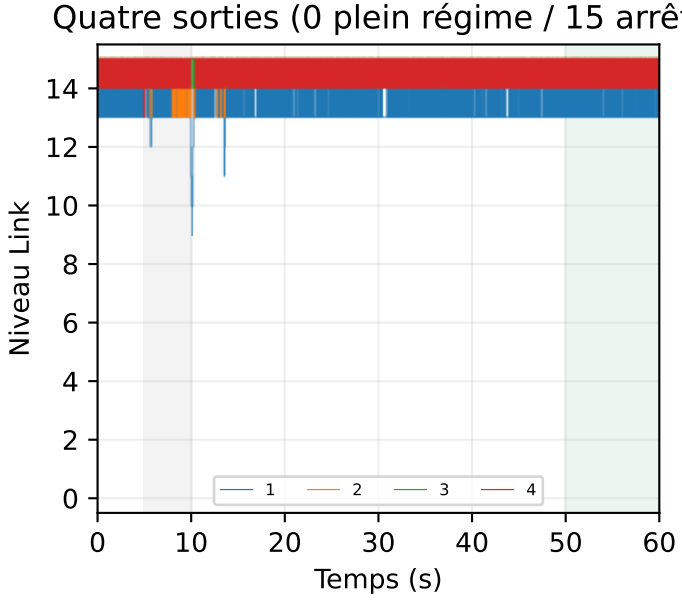
W+Q entre 5 et 10 s, mode déplacement latéral.

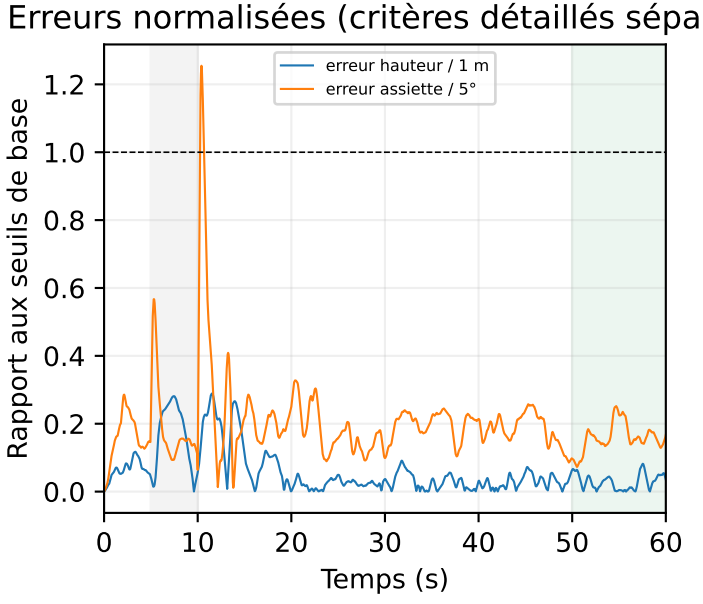












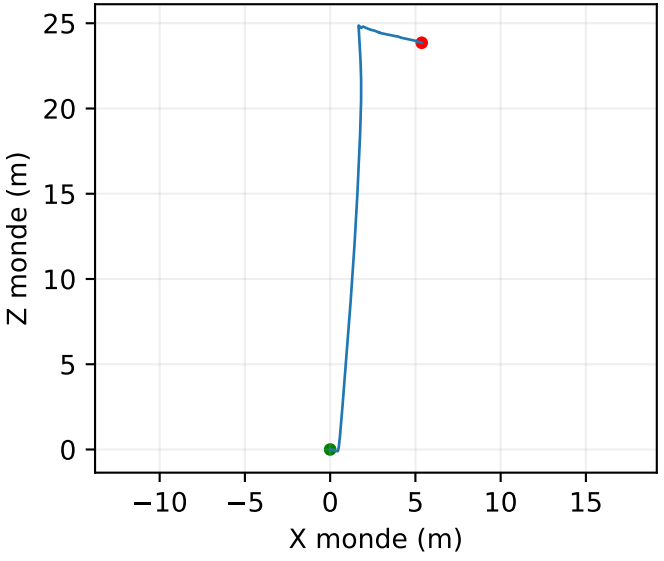
Erreur finale hauteur 0.035 m ; excursion finale 0.149 m ; assiette finale max 0.550° ; freinage 3.900 s. Critères échoués : aucun.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

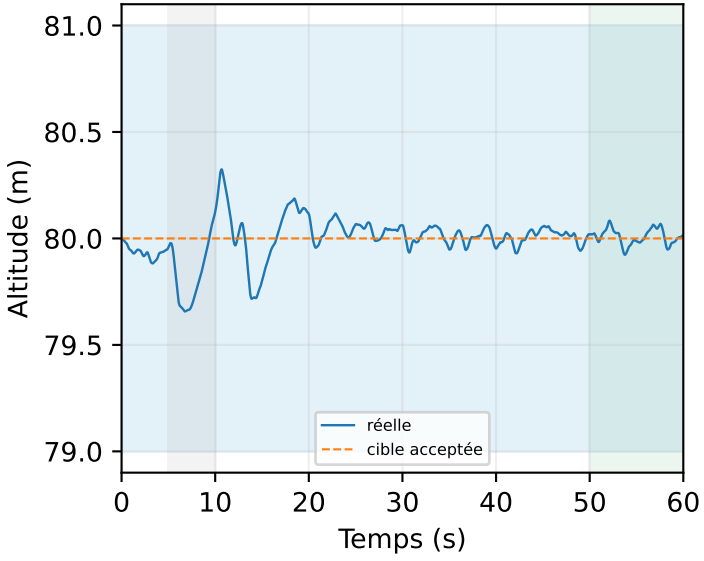
mixed-propellers / limited / brake — PASS

W de 5 à 10 s, relâchement, délai de freinage mesuré sur 15 observations consécutives.

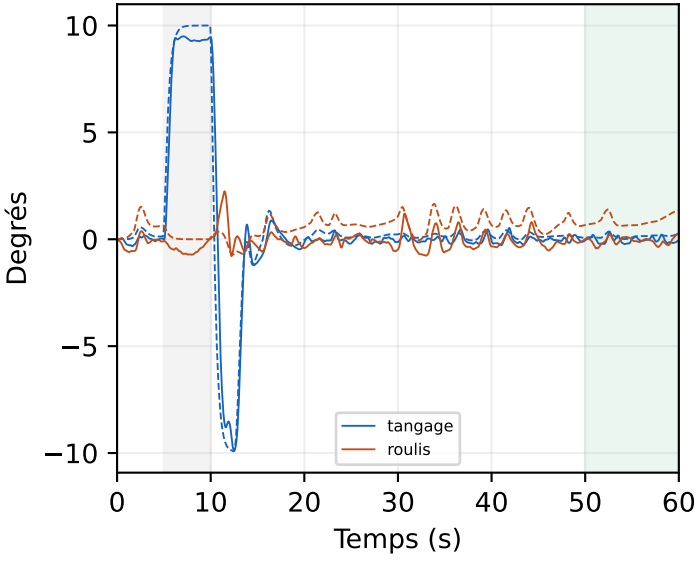
Trajectoire réelle, début vert / fin rouge



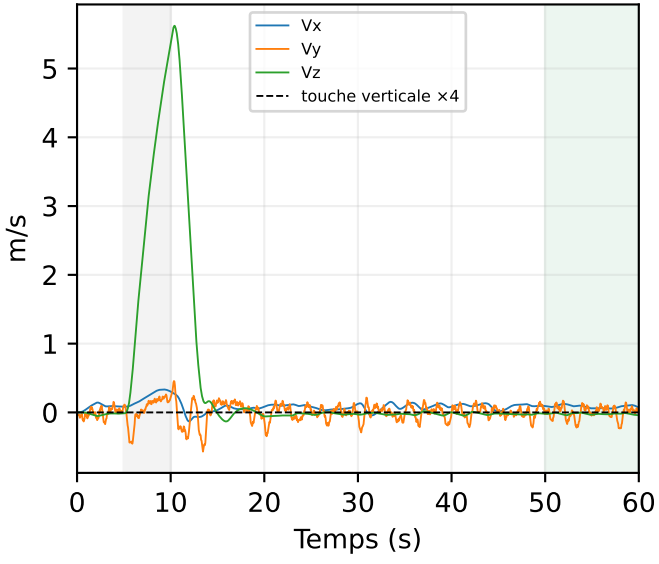
Altitude et bande ±1 m



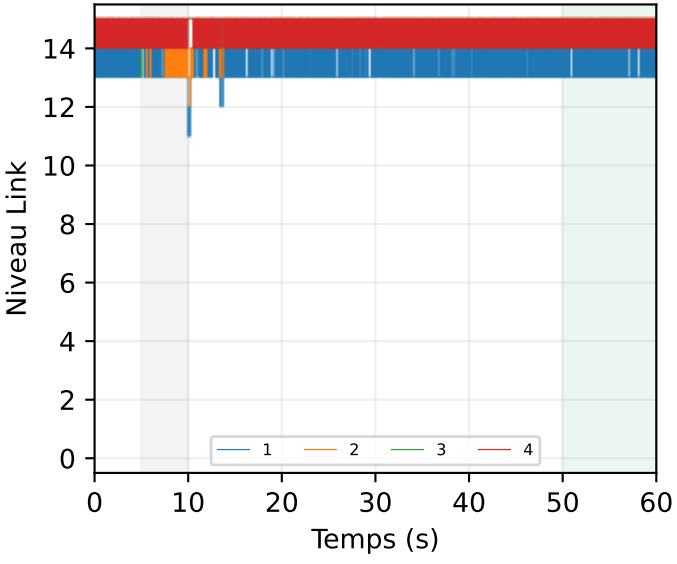
Assiette réelle / cible en pointillé



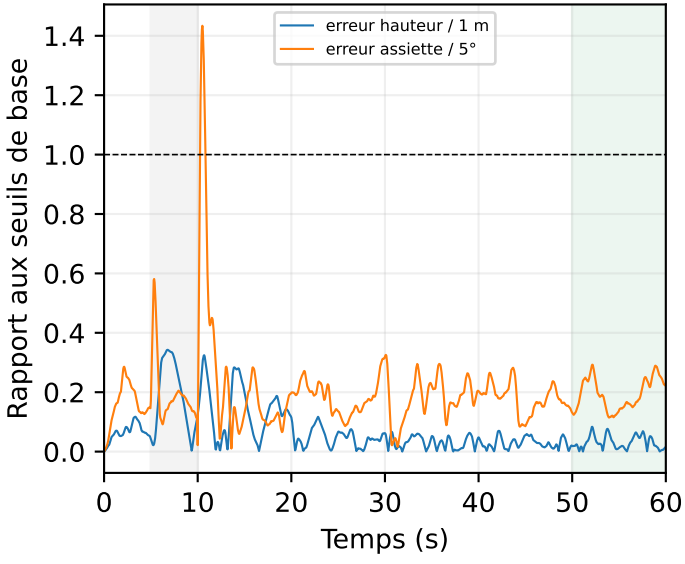
Vitesses et demande verticale



Quatre sorties (0 plein régime / 15 arrêt)



Erreurs normalisées (critères détaillés sépare

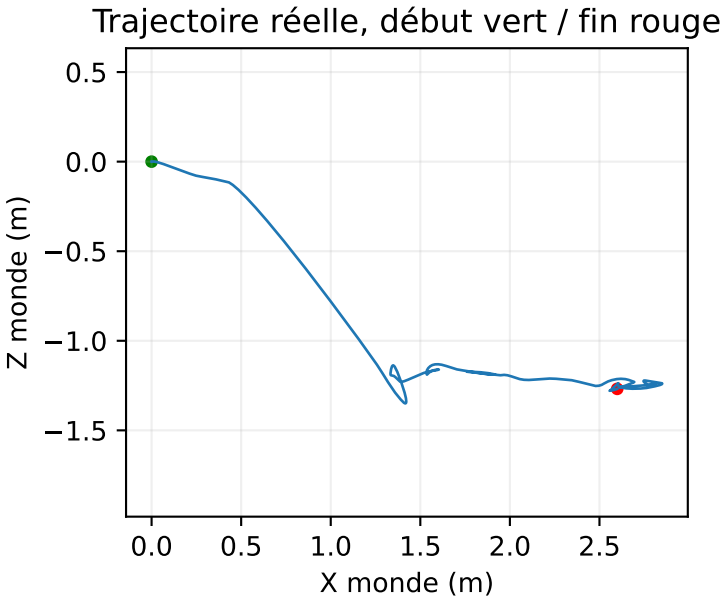


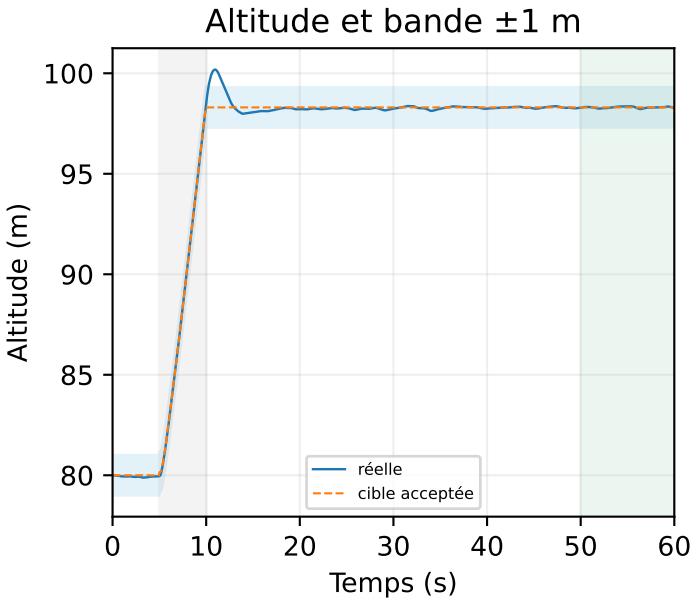
Erreur finale hauteur 0.016 m ; excursion finale 0.160 m ; assiette finale max 0.454° ; freinage 4.000 s. Critères échoués : aucun.

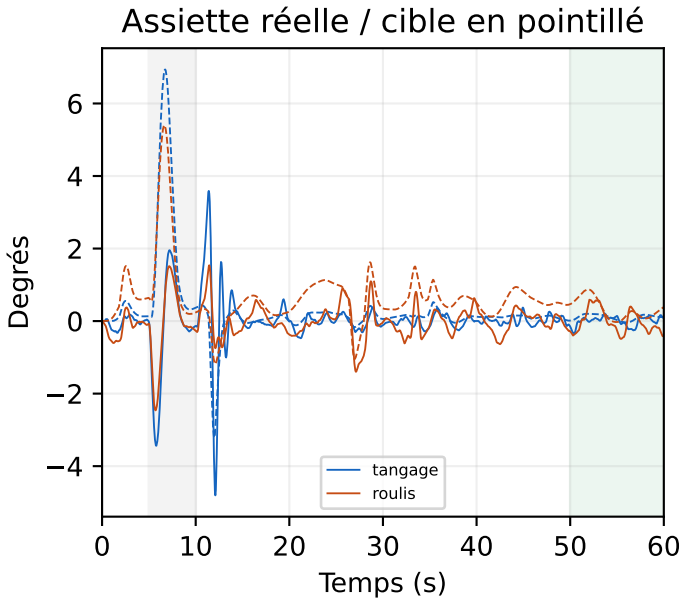
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

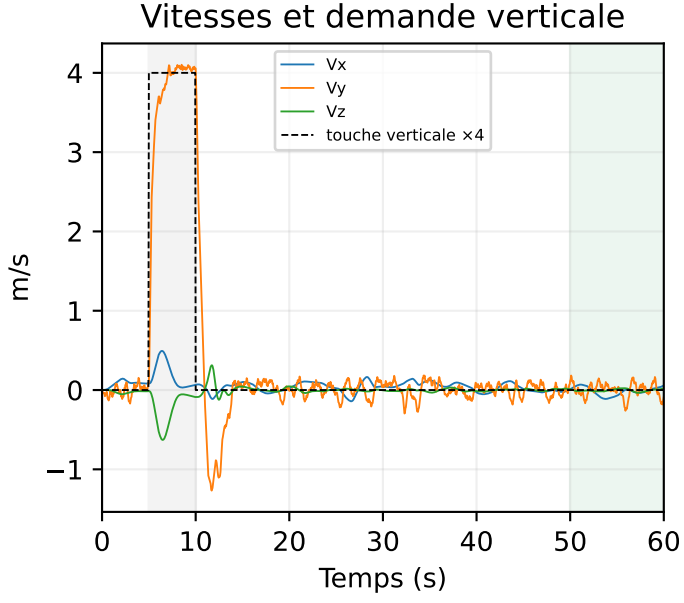
mixed-propellers / limited / climb — PASS

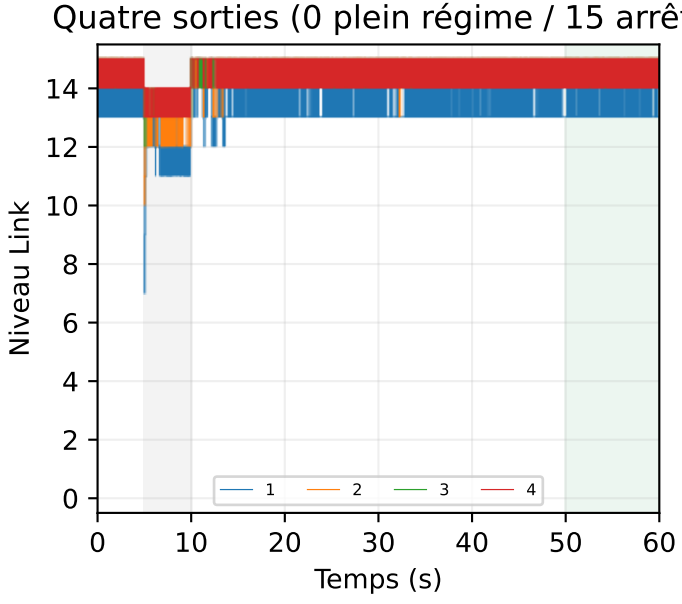
Espace entre 5 et 10 s, puis relâchement et maintien de la hauteur atteinte.

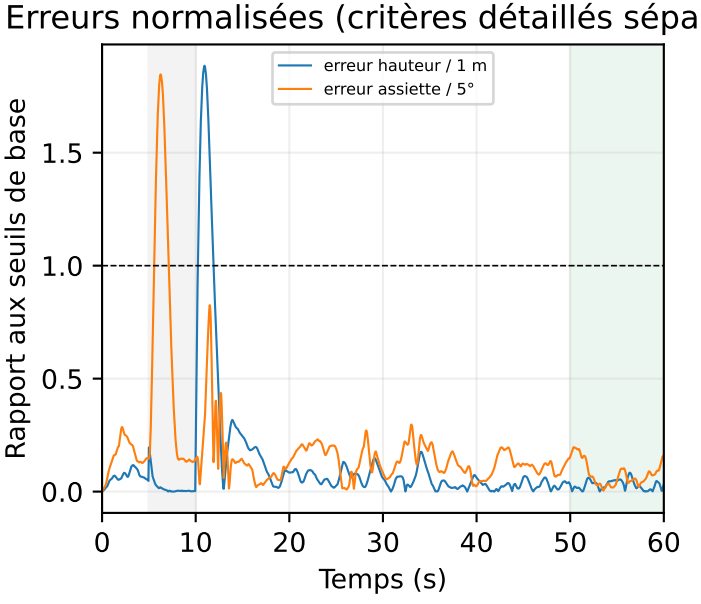












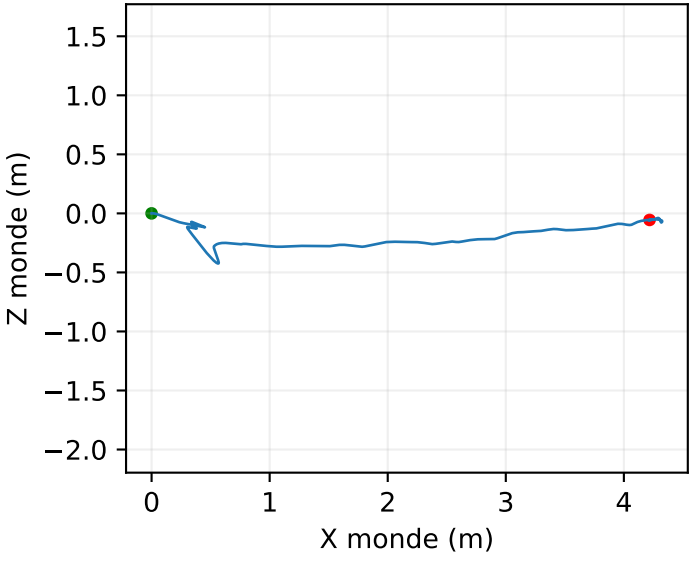
Erreur finale hauteur 0.033 m ; excursion finale 0.138 m ; assiette finale max 0.590° ; freinage — s. Critères échoués : aucun.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

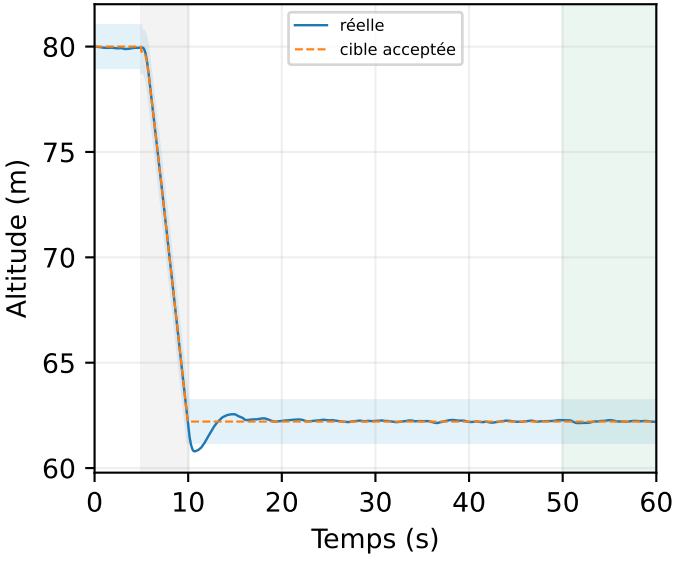
mixed-propellers / limited / descend — PASS

Shift entre 5 et 10 s, puis relâchement et maintien.

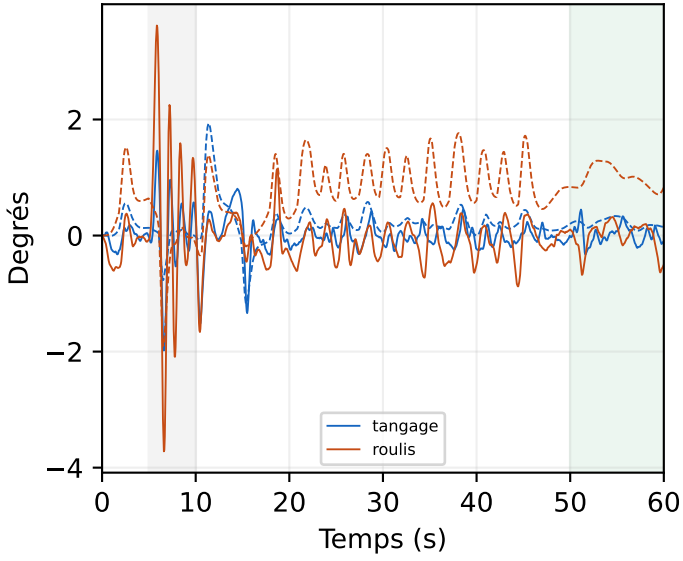
Trajectoire réelle, début vert / fin rouge



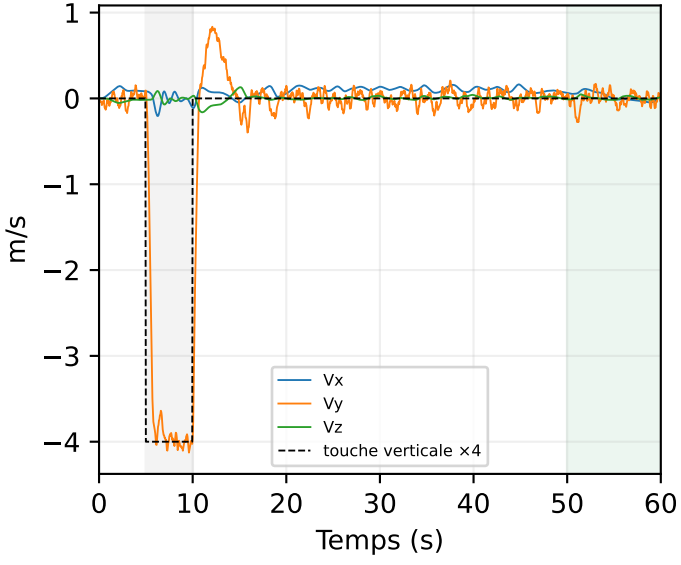
Altitude et bande ±1 m



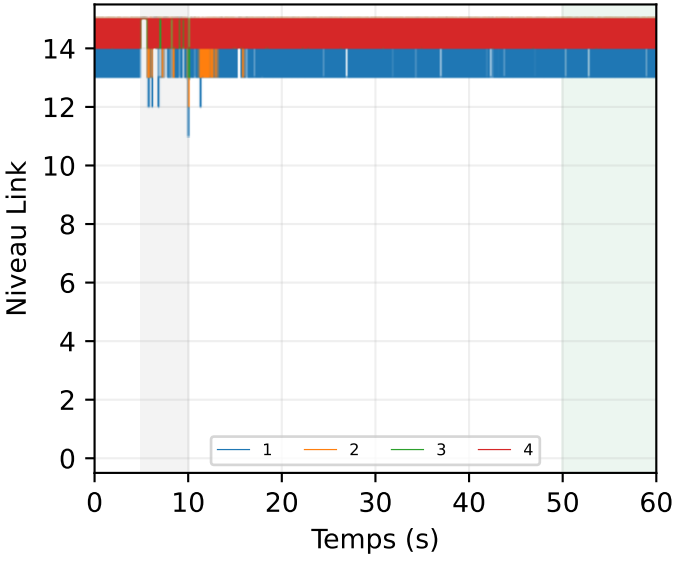
Assiette réelle / cible en pointillé



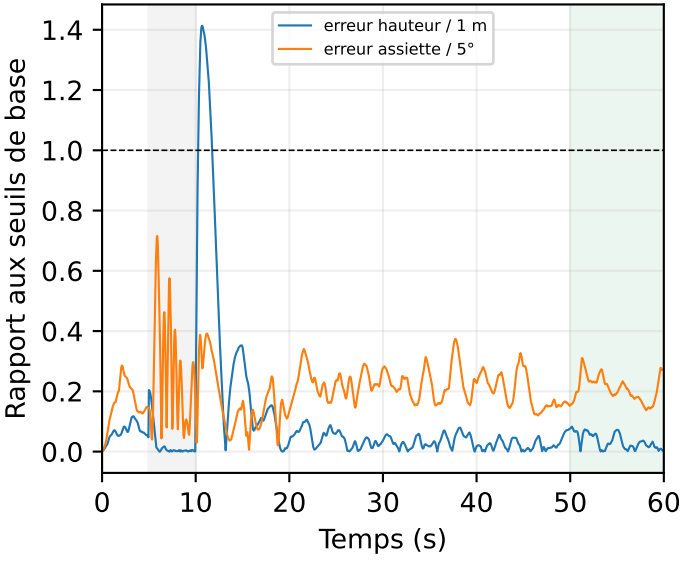
Vitesses et demande verticale



Quatre sorties (0 plein régime / 15 arrêt)



Erreurs normalisées (critères détaillés sépare

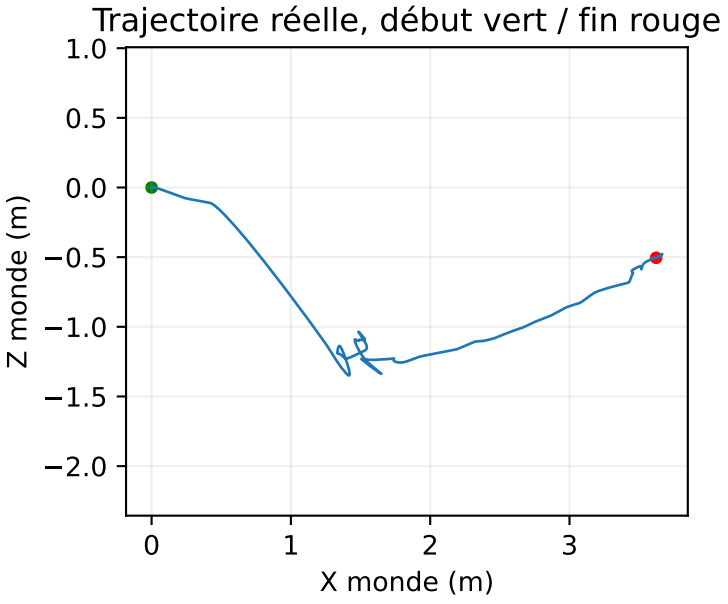


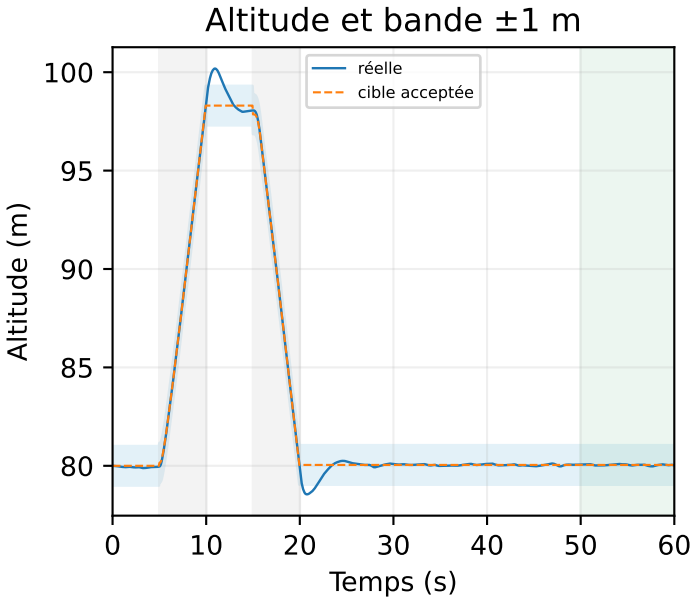
Erreur finale hauteur 0.005 m ; excursion finale 0.156 m ; assiette finale max 0.779° ; freinage — s. Critères échoués : aucun.

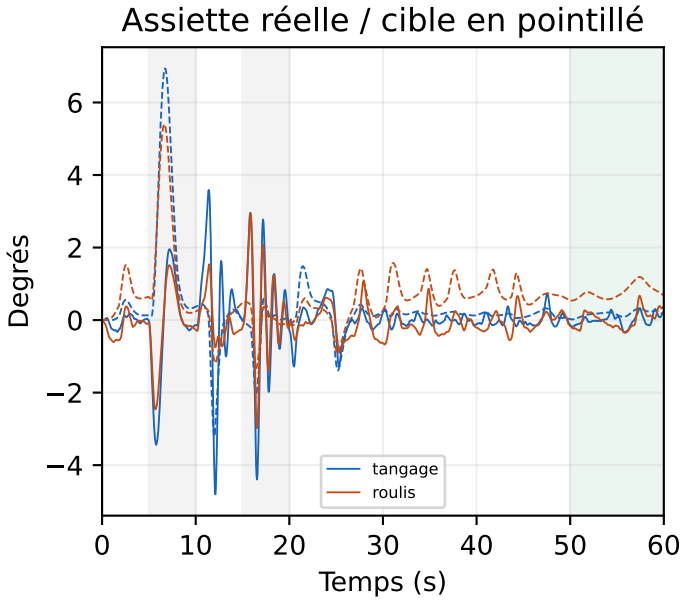
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

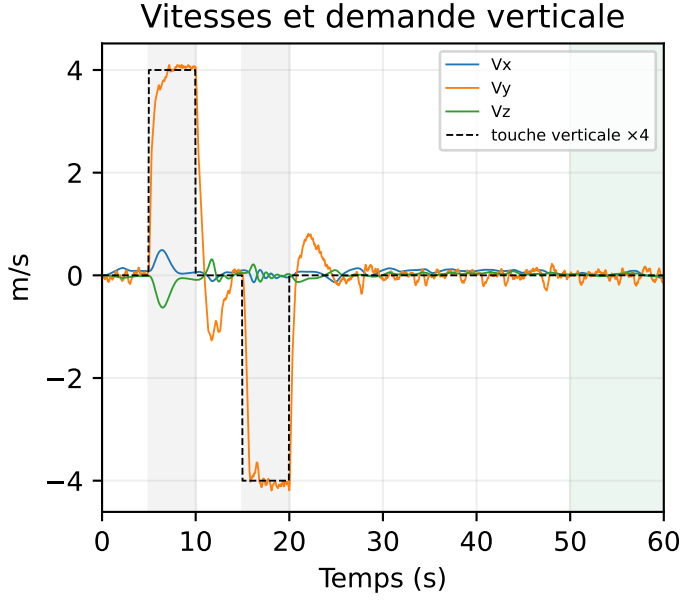
mixed-propellers / limited / vertical-cycle — PASS

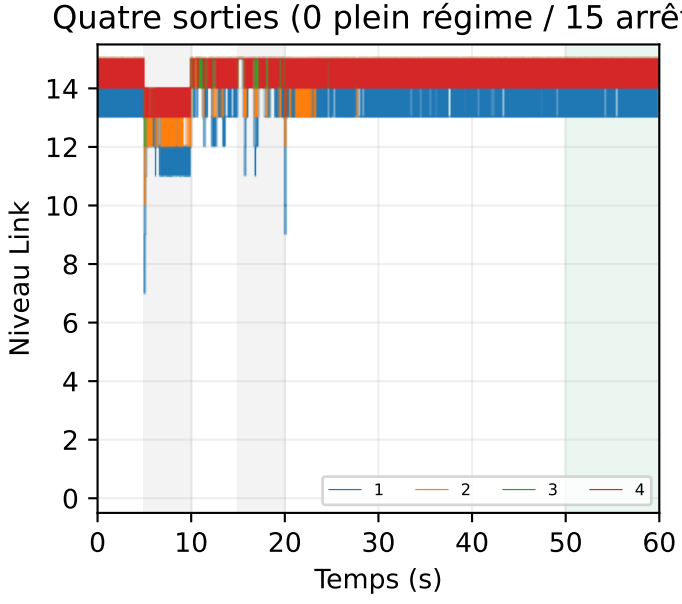
Espace de 5 à 10 s, maintien 10–15 s, Shift 15–20 s, puis maintien.

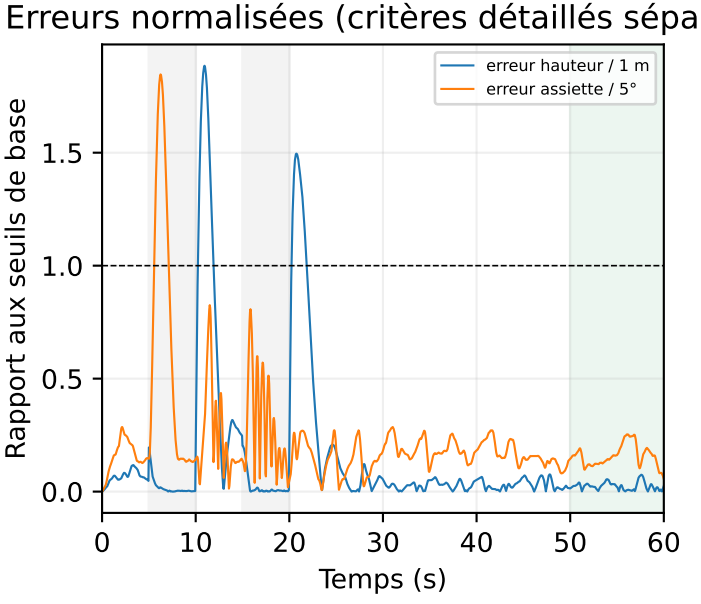










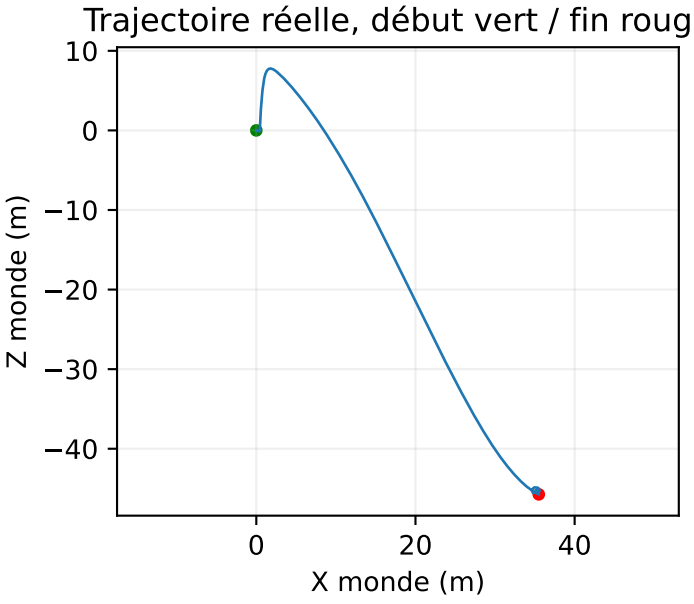


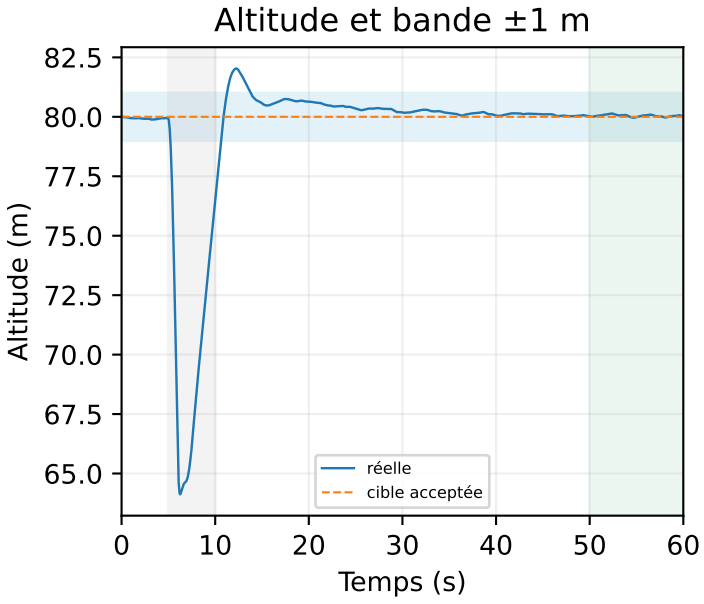
Erreur finale hauteur 0.025 m ; excursion finale 0.130 m ; assiette finale max 0.749° ; freinage — s. Critères échoués : aucun.

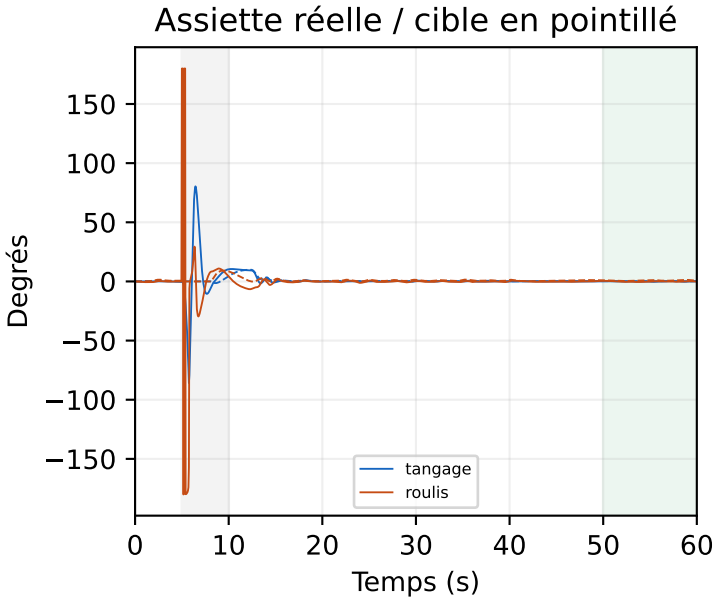
Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.

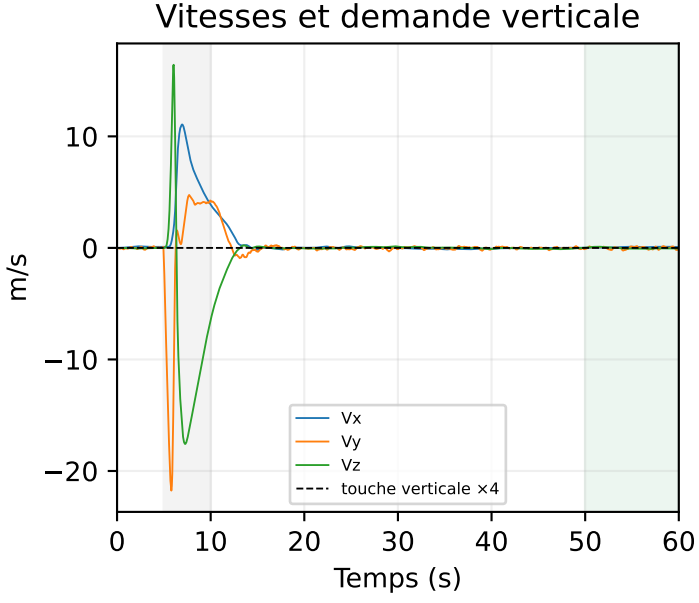
mixed-propellers / limited / recovery-inverted — FAIL

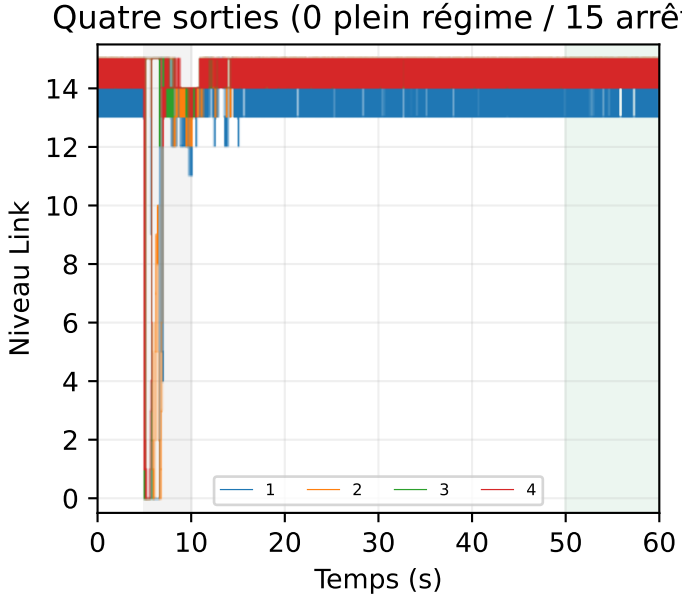
Aucune touche ; inversion instantanée de 180° à 5 s. Perte d’altitude publiée.

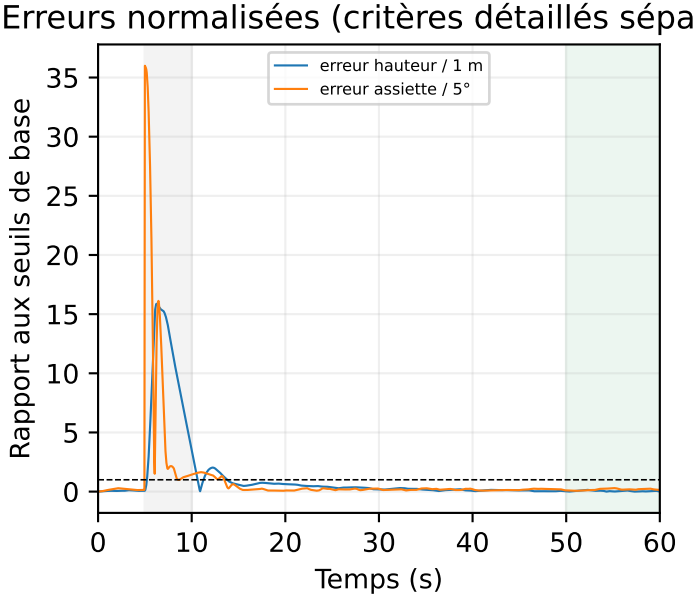












Erreur finale hauteur 0.004 m ; excursion finale 0.186 m ; assiette finale max 0.659° ; freinage — s. Critères échoués : transient_height.

Gris : commande principale. Vert : fenêtre finale de stabilité. Modèle synthétique ; aucune qualification Minecraft.